



REPORT REFERENCE:

IMP7903-1

## **INDUSTRIAL NOISE IMPACT ASSESSMENT**

### **SWIMMING LESSONS & AC UNIT**

British Standard 4142: 2014 +A1:2019

CLIENT:

ET Planning

SITE:

30 Allandale Avenue

London

N3 3PJ

SURVEY DATE:

23<sup>rd</sup> July – 28<sup>th</sup> July 2025

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|----------|---------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>EXECUTIVE SUMMARY.....</b>                                                                                       | <b>4</b>  |
| 1.1      | INSTRUCTION.....                                                                                                    | 4         |
| 1.2      | SCOPE OF REPORT .....                                                                                               | 4         |
| 1.3      | SUMMARY OF REPORT .....                                                                                             | 4         |
| 1.4      | NOISE SOURCES .....                                                                                                 | 4         |
| 1.4.1    | <i>Existing Plant Noise.....</i>                                                                                    | 4         |
| 1.4.2    | <i>Internal Poolside Recorded Noise Levels .....</i>                                                                | 5         |
|          | NOTE: THE HIGHEST OF THE THREE $L_{Aeq,1\text{ HOUR}}$ DB NOISE LEVELS HAVE BEEN USED AS A WORST CASE SCENARIO..... | 5         |
| 1.5      | BS 4142: 2014 +A1:2019 ASSESSMENTS.....                                                                             | 5         |
| 1.6      | SWIMMING LESSONS.....                                                                                               | 5         |
| 1.6.1    | <i>Assessments .....</i>                                                                                            | 5         |
| 1.6.2    | <i>Partial Open Windows.....</i>                                                                                    | 5         |
| 1.7      | CONCLUSIONS.....                                                                                                    | 6         |
| 1.8      | CONCLUSION SUMMARY .....                                                                                            | 6         |
| <b>2</b> | <b>INTRODUCTION .....</b>                                                                                           | <b>7</b>  |
| 2.1      | INSTRUCTION.....                                                                                                    | 7         |
| 2.2      | AMBIENT AND BACKGROUND MEASUREMENTS .....                                                                           | 7         |
| 2.3      | BS 4142: 2014 +A1:2019.....                                                                                         | 7         |
| 2.4      | NATIONAL PLANNING POLICY FRAMEWORK 2024 & NOISE POLICY STATEMENT FOR ENGLAND 2010.....                              | 8         |
| 2.5      | “SIGNIFICANT ADVERSE” AND “ADVERSE” .....                                                                           | 8         |
| 2.6      | NOEL – NO OBSERVED EFFECT LEVEL .....                                                                               | 8         |
| 2.7      | LOAEL – LOWEST OBSERVED ADVERSE EFFECT LEVEL .....                                                                  | 8         |
| 2.8      | SOAEL – SIGNIFICANT OBSERVED ADVERSE EFFECT LEVEL .....                                                             | 8         |
| <b>3</b> | <b>SITE LOCATION .....</b>                                                                                          | <b>9</b>  |
| 3.1      | LOCATION PLAN .....                                                                                                 | 9         |
| <b>4</b> | <b>BACKGROUND NOISE LEVELS.....</b>                                                                                 | <b>10</b> |
| <b>5</b> | <b>APPARATUS.....</b>                                                                                               | <b>11</b> |
| 5.1      | EQUIPMENT CALIBRATION .....                                                                                         | 11        |
| 5.2      | POSITION 1 - 118 NOISE METER s/n 31471 .....                                                                        | 11        |
| <b>6</b> | <b>PLANT CALCULATIONS.....</b>                                                                                      | <b>12</b> |
| 6.1      | EXISTING PLANT NOISE .....                                                                                          | 12        |
| 6.2      | NOISE BREAKOUT CALCULATIONS .....                                                                                   | 12        |
| 6.2.1    | <i>Roof.....</i>                                                                                                    | 12        |
| 6.2.2    | <i>Rear Wall.....</i>                                                                                               | 12        |
| 6.2.3    | <i>Side Wall &amp; Windows Combined .....</i>                                                                       | 13        |
| 6.3      | INPUTTED NOISE LEVELS .....                                                                                         | 13        |
| 6.4      | CALCULATED FAÇADE LEVELS.....                                                                                       | 14        |
| 6.4.1    | <i>Plant Only (BS4142).....</i>                                                                                     | 14        |
| 6.4.2    | <i>Plant &amp; Swimming Lessons.....</i>                                                                            | 14        |
| <b>7</b> | <b>BS4142:2014 +A1:2019 NOISE ASSESSMENT .....</b>                                                                  | <b>15</b> |
| 7.1      | SCOPE OF BRITISH STANDARD 4142: 2014 .....                                                                          | 15        |
| 7.2      | TERMS AND DEFINITIONS .....                                                                                         | 15        |
| 7.3      | ASSESSMENT POSITION .....                                                                                           | 15        |

|           |                                                                                                                        |           |
|-----------|------------------------------------------------------------------------------------------------------------------------|-----------|
| 7.4       | RATING LEVELS (CHARACTER CORRECTION) .....                                                                             | 15        |
| 7.5       | SUBJECTIVE METHOD .....                                                                                                | 16        |
| 7.6       | ASSESSMENT CRITERION .....                                                                                             | 16        |
| 7.7       | ASSESSMENTS .....                                                                                                      | 17        |
| 7.7.1     | Existing Daytime (07:00 – 23:00) .....                                                                                 | 17        |
| 7.8       | ASSESSMENT CONCLUSION .....                                                                                            | 17        |
| <b>8</b>  | <b>SWIMMING LESSONS .....</b>                                                                                          | <b>18</b> |
| 8.1       | INTERNAL POOLSIDE RECORDED NOISE LEVELS .....                                                                          | 18        |
|           | NOTE: THE HIGHEST OF THE THREE $L_{Aeq, 1 \text{ HOUR}}$ dB NOISE LEVELS HAVE BEEN USED AS A WORST CASE SCENARIO. .... | 18        |
| 8.2       | ASSESSMENTS .....                                                                                                      | 18        |
| 8.3       | PARTIAL OPEN WINDOWS .....                                                                                             | 19        |
| 8.3.1     | British Standard 8233: 2014 .....                                                                                      | 19        |
| 8.3.2     | Calculated Internal Noise level .....                                                                                  | 19        |
| <b>9</b>  | <b>NPPF &amp; NPSE .....</b>                                                                                           | <b>20</b> |
| <b>10</b> | <b>CONCLUSIONS .....</b>                                                                                               | <b>21</b> |
| 10.1      | SUMMARY OF REPORT .....                                                                                                | 21        |
| 10.2      | NOISE SOURCES .....                                                                                                    | 21        |
| 10.2.1    | Existing Plant Noise .....                                                                                             | 21        |
| 10.2.2    | Internal Poolside Recorded Noise Levels .....                                                                          | 21        |
|           | NOTE: THE HIGHEST OF THE THREE $L_{Aeq, 1 \text{ HOUR}}$ dB NOISE LEVELS HAVE BEEN USED AS A WORST CASE SCENARIO. .... | 21        |
| 10.3      | BS 4142: 2014 +A1:2019 ASSESSMENTS .....                                                                               | 21        |
| 10.4      | SWIMMING LESSONS .....                                                                                                 | 22        |
| 10.4.1    | Assessments .....                                                                                                      | 22        |
| 10.4.2    | Partial Open Windows .....                                                                                             | 22        |
| 10.5      | CONCLUSIONS .....                                                                                                      | 22        |
| 10.6      | CONCLUSION SUMMARY .....                                                                                               | 22        |
| <b>11</b> | <b>APPENDIX A – RESULTS .....</b>                                                                                      | <b>23</b> |
| 11.1      | 23 <sup>RD</sup> JULY – 24 <sup>TH</sup> JULY 2025 .....                                                               | 23        |
| 11.2      | 24 <sup>TH</sup> JULY – 25 <sup>TH</sup> JULY 2025 .....                                                               | 24        |
| 11.3      | 25 <sup>TH</sup> JULY – 26 <sup>TH</sup> JULY 2025 .....                                                               | 25        |
| 11.4      | 26 <sup>TH</sup> JULY – 27 <sup>TH</sup> JULY 2025 .....                                                               | 26        |
| 11.5      | 27 <sup>TH</sup> JULY – 28 <sup>TH</sup> JULY 2025 .....                                                               | 27        |
| <b>12</b> | <b>APPENDIX B - UNCERTAINTY .....</b>                                                                                  | <b>28</b> |
| 12.1      | UNCERTAINTY BUDGET .....                                                                                               | 28        |
| <b>13</b> | <b>APPENDIX C – WEATHER CONDITIONS .....</b>                                                                           | <b>29</b> |
| <b>14</b> | <b>APPENDIX D – BS 4142:2019+A1:2019 INFORMATION .....</b>                                                             | <b>30</b> |
| 14.1      | MEASUREMENT INFORMATION – LOCATION 1 .....                                                                             | 30        |

## 1 EXECUTIVE SUMMARY

### 1.1 Instruction

Impact Acoustics Ltd has been instructed to undertake a background noise survey to British Standard 4142:2014+A1:2019 to determine the impact of break out noise from swimming lessons at Allandale Avenue, London, N3 3PJ, together with the installation of a single air conditioning units located at the rear of the premises at ground floor level.

### 1.2 Scope of Report

The measurements have been undertaken in accordance with British Standard 4142:2014 +A1:2019 and ISO 1996 – Part 2: 2007 to establish if the proposed site has a demonstrable adverse effect in terms of noise that outweigh the benefits of the development. This report aims to establish the following:

- Existing background noise levels at the Site ( $L_{A90}$ );
- Noise levels from the Swimming Lessons ( $L_{Aeq}$ );
- Impact on the nearest noise sensitive property.
- Mitigation Levels if Required

### 1.3 Summary of Report

Continuous background noise measurements were undertaken between 23rd July and 28th July 2025 at the Assessment Position 1. The daytime background noise levels between 07:00 and 23:00 was found to be  $L_{A90,1 \text{ hour}}$  40 dB. This background noise level has been assessed during daytime periods only as the proposed Swimming Lessons will not operate during night time hours.

|                                                                           |                                          |                    |
|---------------------------------------------------------------------------|------------------------------------------|--------------------|
| Daytime Equivalent Existing<br>Background Noise Levels<br>(07:00 – 23:00) | Measured Levels at Assessment Position 1 | $L_{A90, t}$ 40 dB |
|---------------------------------------------------------------------------|------------------------------------------|--------------------|

### 1.4 Noise Sources

#### 1.4.1 Existing Plant Noise

The following Tables demonstrate the operational noise levels of the Swimming Lessons and the external Condenser unit. These values have been replicated onto the CADNA A 3D model to give a working scenario where suitable mitigation measures can be designed if required.

| Noise Source       | Sound Power Level |
|--------------------|-------------------|
|                    | dB(A)             |
| Fujitsu AOYG14LMCA | 67.0              |

#### 1.4.2 Internal Poolside Recorded Noise Levels

The following noise levels have been recorded within the swimming pool during a typical 3 hour lesson period from 15:00 – 18:00 hrs.

| Noise Source  | Sound Pressure Level | Octave Band Centre Frequency (Hz) |      |      |      |      |      |      |
|---------------|----------------------|-----------------------------------|------|------|------|------|------|------|
|               | dB(A)                | 63                                | 125  | 250  | 500  | 1k   | 2k   | 4k   |
| 15:00 – 16:00 | 66.5                 | 59.7                              | 54.1 | 60.4 | 63.8 | 62.1 | 59.5 | 54.1 |
| 16:00 – 17:00 | 70.7                 | 54.1                              | 52.7 | 60.9 | 66.6 | 67.2 | 64.1 | 57.4 |
| 17:00 – 18:00 | 73.2                 | 52.2                              | 49.9 | 60.1 | 68.7 | 69.7 | 66.9 | 60.6 |

**Note:** The highest of the three  $L_{Aeq\ 1\ HOUR}$  dB noise levels have been used as a worst case scenario.

#### 1.5 BS 4142: 2014 +A1:2019 Assessments

| Rating Level                             | 24 dB                                | 9.1 | The acoustic feature correction is added to the specific noise level |
|------------------------------------------|--------------------------------------|-----|----------------------------------------------------------------------|
| Background Noise Level $L_{A90,1\ hour}$ | 40 dB                                | 8.1 | Modal Background Noise Level                                         |
| Assessment Level                         | -16 dB                               | 11  | The background level is subtracted from the rating level.            |
| Assessment                               | -16 dB                               |     |                                                                      |
| Conclusion                               | The assessment level is 'Low Impact' |     |                                                                      |

#### 1.6 Swimming Lessons

##### 1.6.1 Assessments

| External Residential Assessment | Calculated Façade Level |
|---------------------------------|-------------------------|
| Calculated Level at Residential | 38 dB                   |
| Background Noise Level          | 40 dB                   |
| Excess Over Background          | -2 dB                   |
| Assessment Conclusion           | Low Impact              |

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

##### 1.6.2 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

| External Residential Assessment | Calculated Façade Level   |
|---------------------------------|---------------------------|
| Calculated Level at Residential | 38 dB(A)                  |
| Partially Open Window           | 15 dB(A)                  |
| Calculated internal Noise level | 23 dB(A)                  |
| BS8233: 2014 criteria           | $L_{Aeq\ 16\ Hour}$ 30 dB |
| Excess Over Criteria            | -7 dB                     |
| Assessment Conclusion           | Low Impact                |

## **1.7 Conclusions**

The operation of the condenser and the break out noise have been assessed to establish if the swimming lessons will have a demonstrable adverse effect in terms of noise that outweigh the benefits of the development. Measurements have been undertaken in accordance with British Standard 4142: 2014 +A1:2019 and ISO 1996 – Part 2: 2017. This report has established the existing background noise levels at the closest noise sensitive façade to the site and the assessment of the impact of the site operation on nearby residential properties to the east of the site. The resulting emissions from the site running on a worst case scenario show no conflict with ‘Low Impact’ criteria and give a strong indication that complaint and impact on the local amenity is unlikely.

## **1.8 Conclusion Summary**

National Planning Policy Framework 2024 suggests that planning permission should be granted unless any adverse impacts of doing so would significantly and demonstrably outweigh the benefits, when assessed against the policies in the framework taken as a whole, or specific policies in the framework indicate the application should be restricted.

Based on the calculations and assessments made within this report it is the professional opinion of Impact Acoustics Ltd that the proposed development can demonstrate compliance with the National Planning Policy Framework 2024, NPPF & NPSE and that, with regards to sound, planning permission can be granted.

## **2 INTRODUCTION**

### **2.1 Instruction**

Impact Acoustics Ltd has been instructed to undertake a background noise survey to British Standard 4142:2014+A1:2019 to determine the impact of break out noise from swimming lessons at Allandale Avenue, London, N3 3PJ, together with the installation of a single air conditioning units located at the rear of the premises at ground floor level.

The purpose of this assessment is to ensure the acoustic protection of noise sensitive premises closest to the swimming pool. Noise sensitive premises are not restricted to residential dwellings as offices can be affected by unwanted external noise. These have been established as the residential properties to either side of the site.

### **2.2 Ambient and Background Measurements**

External noise levels are to be recorded over, what has been considered for the site, an average / typical time period in order to assess the daytime noise levels. Levels have been recorded over more than one day in order to ensure the uncertainty of measurement aspects of BS 4142: 2014 +A1:2019 have been satisfied and that the data recorded is representative for the purpose of a robust assessment.

### **2.3 BS 4142: 2014 +A1:2019**

British Standard 4142: 2014 +A1:2019 is to be adopted for the basis of this background noise level assessment. A BS 4142: 2014 +A1:2019 noise assessment will be carried out in order to demonstrate the acoustic impact the Condenser could have on the nearest affected premises and make suitable recommendations in order to demonstrate this unit will not have a significant and demonstrable adverse impact on the nearest noise sensitive premises in accordance with the National Planning Policy Framework, once remedial works are completed.

## **2.4 National Planning Policy Framework 2024 & Noise Policy Statement for England 2010**

References and evaluations are to be made to the National Planning Policy Framework 2024 (NPPF) and the Noise Policy Statement for England 2010 (NPSE). The purpose of this document is to include all aspects of environmental noise within assessments i.e. environmental noise, neighbour noise and neighbourhood noise. Noise is to be considered alongside other relevant issues relating to the site and should not be considered in isolation, according to the NPSE.

There are several key phrases within the NPSE aims and these are discussed below.

### **2.5 “Significant adverse” and “adverse”**

*There are two established concepts from toxicology that are currently being applied to noise impacts, for example, by the World Health Organisation. They are:*

### **2.6 NOEL – No Observed Effect Level**

*This is the level below which no effect can be detected. In simple terms, below this level, there is no detectable effect on health and quality of life due to the noise.*

### **2.7 LOAEL – Lowest Observed Adverse Effect Level**

*This is the level above which adverse effects on health and quality of life can be detected.*

*Extending these concepts for the purpose of this NPSE leads to the concept of a significant observed adverse effect level.*

### **2.8 SOAEL – Significant Observed Adverse Effect Level**

*This is the level above which significant adverse effects on health and quality of life occur.*

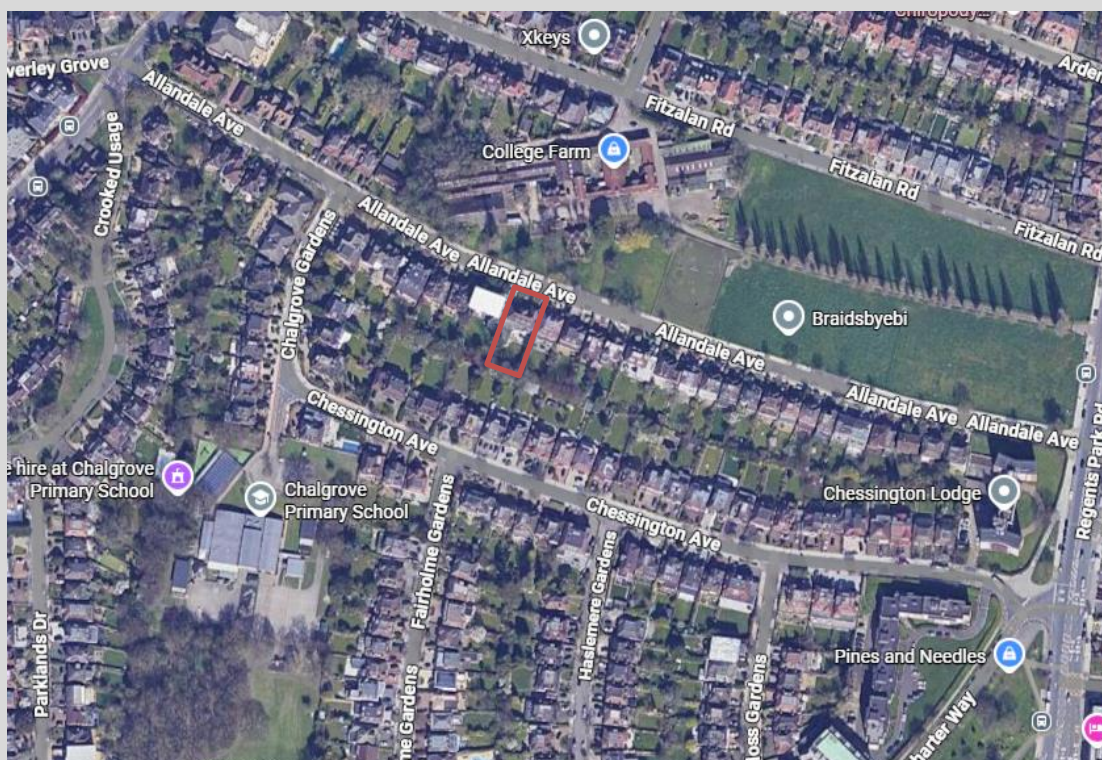
*It is not possible to have a single objective noise-based measure that defines SOAEL that is applicable to all sources of noise in all situations. Consequently, the SOAEL is likely to be different for different noise sources, for different receptors and at different times. It is acknowledged that further research is required to increase our understanding of what may constitute a significant adverse impact on health and quality of life from noise. However, not having specific SOAEL values in the NPSE provides the necessary policy flexibility until further evidence and suitable guidance is available.*



### 3 SITE LOCATION

#### 3.1 Location Plan

- Yellow – Background Measurement Position
- Orange NSP (Noise Sensitive Premises)



4 BACKGROUND NOISE LEVELS

Continuous background noise measurements were undertaken between 23<sup>rd</sup> July and 28<sup>th</sup> July 2025 at the Assessment Position 1. The daytime background noise levels between 07:00 and 23:00 was found to be  $L_{A90,1 \text{ hour}}$  40 dB. This background noise level has been assessed during daytime periods only as the Swimming Lessons will not operate during night time hours.

|                                                                           |                                          |                    |
|---------------------------------------------------------------------------|------------------------------------------|--------------------|
| Daytime Equivalent Existing<br>Background Noise Levels<br>(07:00 – 23:00) | Measured Levels at Assessment Position 1 | $L_{A90, t}$ 40 dB |
|---------------------------------------------------------------------------|------------------------------------------|--------------------|

## 5 APPARATUS

### 5.1 Equipment Calibration

The equipment was calibrated using a sound pressure level of 114.0 dB at an octave band centre frequency of 1000Hz with reference to  $2 \times 10^{-5} \text{ Nm}^{-2}$  before and after the tests and the equipment set to have no inaccuracy greater than 0.2 dB.

All the following equipment was calibrated in accordance with the laboratory accreditation requirements of the United Kingdom Accreditation Service (UKAS) on the following dates. Calibration schedules are implemented within Impact Acoustics Ltd in accordance with UKAS directive LAB 23.

### 5.2 Position 1 - 118 NOISE METER s/n 31471

| Description                         | Make     | Type | Serial No. | Calibration Intervals | Cert No     | Last Calibrated | Next Due Calibration |
|-------------------------------------|----------|------|------------|-----------------------|-------------|-----------------|----------------------|
| Integrated Sound Level Meter        | Norsonic | 118  | 31471      | 2 YEARS               | TCRT23-1042 | 15.01.2024      | 15.01.2026           |
| 12.5mm Microphone (with windshield) | GRAS     | 40AF | 114650     | 2 YEARS               |             | 15.01.2024      | 15.01.2026           |
| Microphone Pre – Amplifier          | Norsonic | 1206 | 30327      | 2 YEARS               |             | 15.01.2024      | 15.01.2026           |

The noise meter was calibrated before and after the assessment period and found to be within the tolerance of the manufacturer's guidance. Full Calibration certificates are available upon request.

## 6 PLANT CALCULATIONS

### 6.1 Existing Plant Noise

The following Tables demonstrate the operational noise levels of the Swimming Lessons and the external Condenser unit. These values have been replicated onto the CADNA A 3D model to give a working scenario where suitable mitigation measures can be designed if required.

| Noise Source       | Sound Power Level |
|--------------------|-------------------|
|                    | dB(A)             |
| Fujitsu AOYG14LMCA | 67.0              |

### 6.2 Noise Breakout Calculations

#### 6.2.1 Roof

| Roof                            | dB(A) | Octave Band Centre Frequency (Hz) |      |      |      |      |      |      |
|---------------------------------|-------|-----------------------------------|------|------|------|------|------|------|
|                                 |       | 63                                | 125  | 250  | 500  | 1    | 2    | 4    |
| Internal Noise Level            | 73.2  | 52.2                              | 49.9 | 60.1 | 68.7 | 69.7 | 66.9 | 60.6 |
| R - Reduction Value             |       | 27.6                              | 36.5 | 38.1 | 45.5 | 52.2 | 51   | 49.8 |
| S - Surface Area                | 58.8  |                                   |      |      |      |      |      |      |
| r - Distance (m)                | 1     |                                   |      |      |      |      |      |      |
| 10xLog(s)                       | 17.7  | 17.7                              | 17.7 | 17.7 | 17.7 | 17.7 | 17.7 | 17.7 |
| 20xLog r                        | 0.0   | 0.0                               | 0.0  | 0.0  | 0.0  | 0.0  | 0.0  | 0.0  |
| External Calculated Level (Lw2) | 46.6  | 47.3                              | 36.1 | 44.7 | 45.9 | 40.1 | 38.6 | 33.5 |

#### 6.2.2 Rear Wall

| Rear Wall                       | dB(A) | Octave Band Centre Frequency (Hz) |      |      |      |      |      |      |
|---------------------------------|-------|-----------------------------------|------|------|------|------|------|------|
|                                 |       | 63                                | 125  | 250  | 500  | 1    | 2    | 4    |
| Internal Noise Level            | 73.2  | 52.2                              | 49.9 | 60.1 | 68.7 | 69.7 | 66.9 | 60.6 |
| R - Reduction Value             |       | 30.6                              | 39.8 | 39.3 | 48.7 | 55.9 | 55.1 | 46.8 |
| S - Surface Area                | 13    |                                   |      |      |      |      |      |      |
| r - Distance (m)                | 1     |                                   |      |      |      |      |      |      |
| 10xLog(s)                       | 11.1  | 11.1                              | 11.1 | 11.1 | 11.1 | 11.1 | 11.1 | 11.1 |
| 20xLog r                        | 0.0   | 0.0                               | 0.0  | 0.0  | 0.0  | 0.0  | 0.0  | 0.0  |
| External Calculated Level (Lw2) | 37.6  | 37.8                              | 26.3 | 36.9 | 36.1 | 29.9 | 28.0 | 30.0 |

### 6.2.3 Side Wall & Windows Combined

| Front & Side Walls & Glazing    | dB(A) | Octave Band Centre Frequency (Hz) |      |      |      |      |      |      |
|---------------------------------|-------|-----------------------------------|------|------|------|------|------|------|
|                                 |       | 63                                | 125  | 250  | 500  | 1    | 2    | 4    |
| Internal Noise Level            | 73.2  | 52.2                              | 49.9 | 60.1 | 68.7 | 69.7 | 66.9 | 60.6 |
| R - Reduction Value             |       | 24.0                              | 32.4 | 30.3 | 35.0 | 39.5 | 44.7 | 37.7 |
| S - Surface Area                | 48    |                                   |      |      |      |      |      |      |
| r - Distance (m)                | 1     |                                   |      |      |      |      |      |      |
| 10xLog(s)                       | 16.8  | 16.8                              | 16.8 | 16.8 | 16.8 | 16.8 | 16.8 | 16.8 |
| 20xLog r                        | 0.0   | 0.0                               | 0.0  | 0.0  | 0.0  | 0.0  | 0.0  | 0.0  |
| External Calculated Level (Lw2) | 56.3  | 50.0                              | 39.4 | 51.6 | 55.5 | 52.0 | 44.0 | 44.7 |

### 6.3 Inputted Noise Levels

Based on the above calculations, the following noise levels have been inputted into CADNA A.

| Sound Levels (local)                                                                                                                                                                                                                       |                        |      |                      |      |      |      |      |      |      |      |      |      |      |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------|----------------------|------|------|------|------|------|------|------|------|------|------|--------|
| <input type="button" value="OK"/> <input type="button" value="Cancel"/> <input type="button" value="Copy..."/> <input type="button" value="Font..."/> <input type="button" value="Adjust Col. Width"/> <input type="button" value="Help"/> |                        |      |                      |      |      |      |      |      |      |      |      |      |      |        |
| Name                                                                                                                                                                                                                                       | ID                     | Type | Oktave Spectrum (dB) |      |      |      |      |      |      |      |      |      |      | Source |
|                                                                                                                                                                                                                                            |                        |      | Weight.              | 31.5 | 63   | 125  | 250  | 500  | 1000 | 2000 | 4000 | 8000 | A    | lin    |
| RearWall                                                                                                                                                                                                                                   | RearWall               | Li   |                      | 0.0  | 37.8 | 26.3 | 36.9 | 36.1 | 29.9 | 28.0 | 30.0 | 0.0  | 37.6 | 42.6   |
| Side&FrontWall&Windows                                                                                                                                                                                                                     | Side&FrontWall&Windows | Li   |                      | 0.0  | 50.0 | 39.4 | 51.6 | 55.5 | 52.0 | 44.0 | 44.7 | 0.0  | 56.2 | 59.1   |
| Roof                                                                                                                                                                                                                                       | Roof                   | Li   |                      | 0.0  | 47.3 | 36.1 | 44.7 | 45.9 | 40.1 | 38.6 | 33.5 | 0.0  | 46.6 | 51.6   |

Inputted Noise Levels



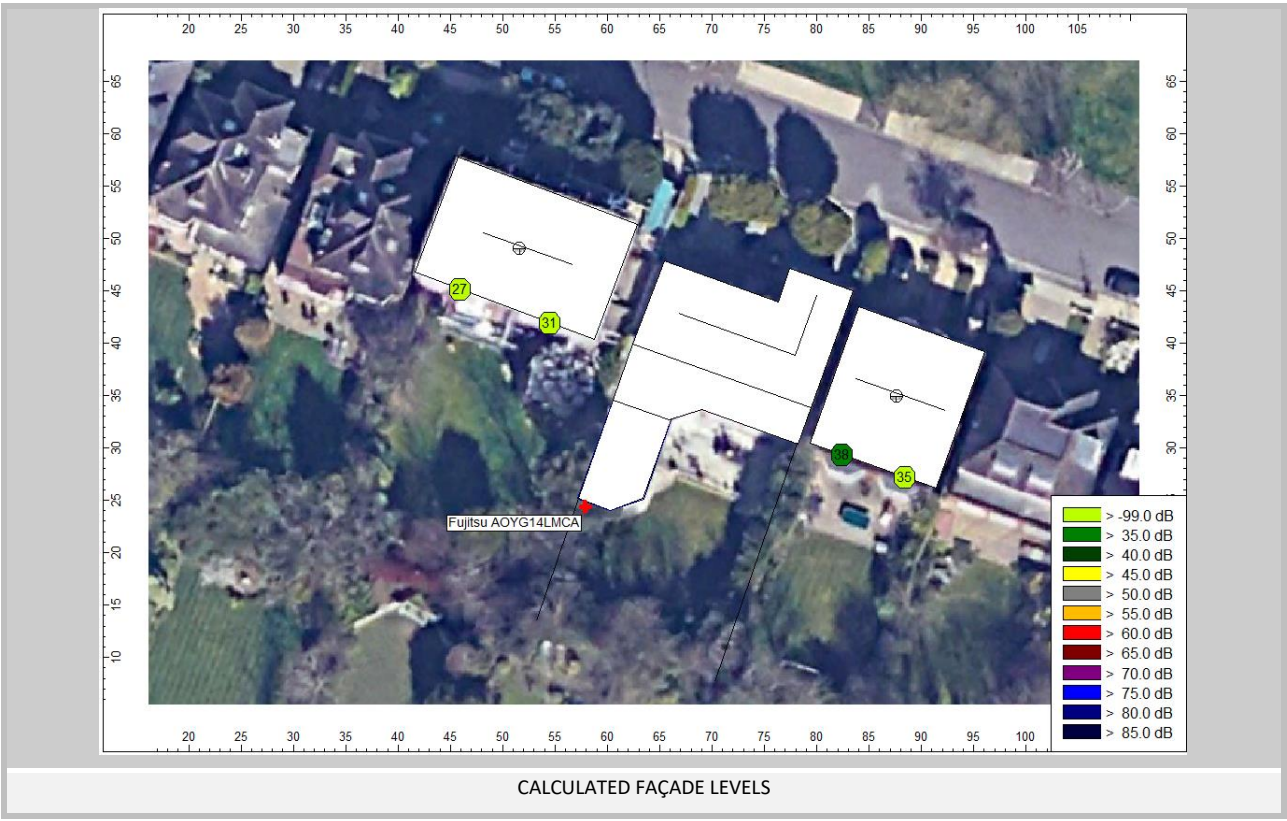
6.4 Calculated Façade Levels

The following demonstrates the above noise levels replicated on the 3D model.

6.4.1 Plant Only (BS4142)



6.4.2 Plant & Swimming Lessons



## 7 BS4142:2014 +A1:2019 NOISE ASSESSMENT

### 7.1 Scope of British Standard 4142: 2014

In the assessment of the existing surrounding commercial premises, consideration has been given to the scope of British Standard 4142:2014+A1:2019, which in section 1, details applicability of this standard to rating assessing sound of an industrial and/or commercial nature. It is considered appropriate that both the background noise levels, and the rating noise levels obtained fall within the scope of British Standard 4142:2014+A1:2019 by using outdoor sound levels to assess the effect of sound on local residents.

### 7.2 Terms and Definitions

| Symbol             | Term                | Definition                                                                                                                                                              |
|--------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AP                 | Assessment Position | Position externally at the façade property under investigation at which the assessment is undertaken which is usually 1m from the 1 <sup>st</sup> floor bedroom window. |
| EP                 | Equivalent Position | Position at which the background noise levels are measured if there is no access to the assessment position or if source under investigation is audible.                |
| $L_s = L_{Aeq, T}$ | Specific Level      | The average continuous equivalent sound pressure level of the source at the assessment position.                                                                        |
| $L_{Ar, Tr}$       | Rating Level        | The average continuous equivalent sound pressure level of the source at the assessment position with a correction to account for the characteristic features.           |
| $L_r = L_{Aeq, T}$ | Residual Level      | The average continuous equivalent sound pressure level at the assessment position without the source operating.                                                         |
| $L_{A90, T}$       | Background Level    | The sound pressure level that is not exceeded 90% of the time at the assessment position.                                                                               |
| $L_a = L_{Aeq, T}$ | Ambient Level       | The totally encompassing sound at the assessment position including the residual and specific noise.                                                                    |

### 7.3 Assessment Position

The assessment position 1 was established as the proposed residential accommodation premises to the south of the Pitter Commercial Site.

### 7.4 Rating Levels (Character Correction)

It is appropriate to add a character correction where there is a new source that cannot be measured in line with BS 4142:2014+A1:2019. There are 3 methods for approaching this.

- Subjective method
- Objective method (for tonality)
- Reference method

## 7.5 Subjective Method

The subjective method establishes a rating penalty that is added to the specific noise level if any of the following is present at the assessment position. If a tone is expected to be present a character correction of 0 dB to 6 dB is added depending on how perceptible it is at noise sensitive locations.

| BS 4142:2014+A1:2019 – Section 9.2<br>Subjective Method | Perceptibility to noise sensitive facades | Correction |
|---------------------------------------------------------|-------------------------------------------|------------|
| Tonality<br>Ranging from not tonal to prominently tonal | Not tonal                                 | +0         |
|                                                         | Just perceptible                          | +2         |
|                                                         | Clearly perceptible                       | +4         |
|                                                         | Highly perceptible                        | +6         |

If the source is expected to be impulsive a character correction of 0 dB to 9 dB is added depending on how perceptible it is at noise sensitive locations.

| BS 4142:2014+A1:2019 – Section 9.2<br>Subjective Method                             | Perceptibility to noise sensitive facades | Correction |
|-------------------------------------------------------------------------------------|-------------------------------------------|------------|
| Impulsivity<br>Considering both the rapidity and any overall change in sound levels | Not impulsive                             | +0         |
|                                                                                     | Just perceptible                          | +3         |
|                                                                                     | Clearly perceptible                       | +6         |
|                                                                                     | Highly perceptible                        | +9         |

When the sound features are neither tonal nor impulsive, a character correction of +3 is added for the readily distinctive quality against the acoustic environment or for the intermittency of the source.

| BS 4142:2014+A1:2019 – Section 9.2<br>Subjective Method | Perceptibility to noise sensitive facades | Correction |
|---------------------------------------------------------|-------------------------------------------|------------|
| Readily Distinctive                                     | Is not present                            | +0         |
|                                                         | Is present                                | +3         |
| Intermittency                                           | Is not present                            | +0         |
|                                                         | Is present                                | +3         |

## 7.6 Assessment Criterion

The significance of the resulting noise on the residential property depends on the margin by which it exceeds the background noise levels. British Standard 4142:2014+A1:2019 provides the following guidance within section 11.

| Difference | Assessment of Impact                       |
|------------|--------------------------------------------|
| +10 dB     | Indication of a significant adverse impact |
| +5 dB      | Indication of an adverse impact            |
| +0 dB      | Indication of low impact                   |



## 7.7 Assessments

### 7.7.1 Existing Daytime (07:00 – 23:00)

| Item                                            | Calculation                          | Clause | Commentary                                                           |
|-------------------------------------------------|--------------------------------------|--------|----------------------------------------------------------------------|
| Specific Noise Level $L_{Aeq,1 \text{ hour}}$   | 17 dB                                | 7      | Calculated using ISO 9613:1996 <sup>[3]</sup> .                      |
| Tonal                                           | +2 dB                                | 8.1    | Tonality Characteristic                                              |
| Intermittency                                   | +3 dB                                | 8.1    | Intermittency Characteristic                                         |
| Rating Level                                    | 24 dB                                | 9.1    | The acoustic feature correction is added to the specific noise level |
| Background Noise Level $L_{A90,1 \text{ hour}}$ | 40 dB                                | 8.1    | Modal Background Noise Level                                         |
| Assessment Level                                | -16 dB                               | 11     | The background level is subtracted from the rating level.            |
| Assessment                                      | -16 dB                               |        |                                                                      |
| Conclusion                                      | The assessment level is 'Low Impact' |        |                                                                      |

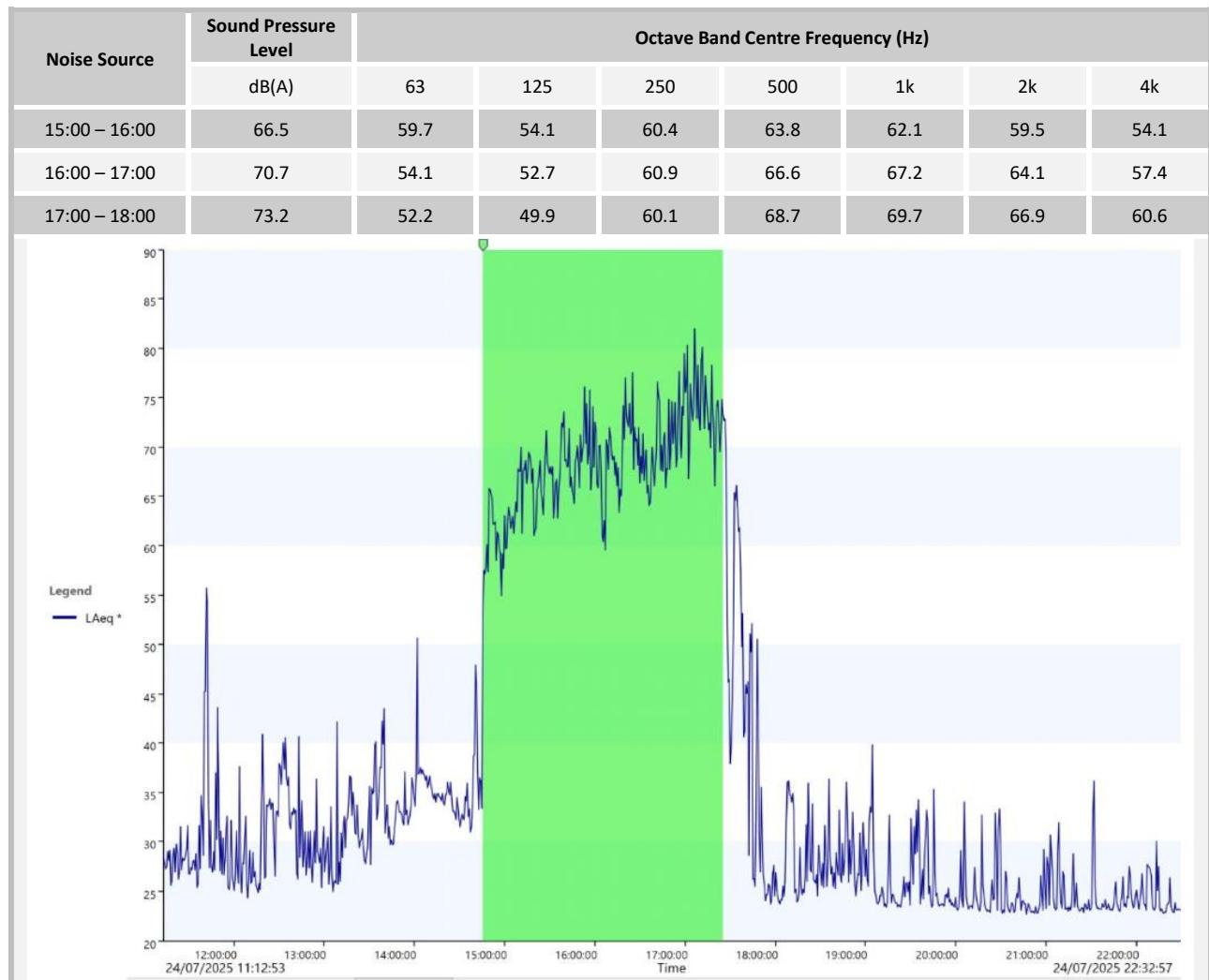
## 7.8 Assessment Conclusion

It can be seen from the above assessments that an assessment conclusion of Low Impact could be expected. Therefore, no further mitigation measures would be required to the plant at this stage.

## 8 SWIMMING LESSONS

### 8.1 Internal Poolside Recorded Noise Levels

The following noise levels have been recorded within the swimming pool during a typical 3 hour lesson period from 15:00 – 18:00 hrs.



**Note:** The highest of the three  $L_{Aeq, 1 \text{ HOUR}}$  dB noise levels have been used as a worst case scenario.

### 8.2 Assessments

| External Residential Assessment | Calculated Façade Level |
|---------------------------------|-------------------------|
| Calculated Level at Residential | 38 dB                   |
| Background Noise Level          | 40 dB                   |
| Excess Over Background          | -2 dB                   |
| Assessment Conclusion           | Low Impact              |

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

### 8.3 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

#### 8.3.1 British Standard 8233: 2014

Calculations are to be made in accordance with BS 8233: 2014 Sound Insulation and Noise Reduction for Buildings Code of Practice.

BS 8233: 2014 set the following parameters as target levels that should be designed to within rooms such as Living Rooms and Bedrooms.

| Indoor ambient noise levels for dwellings |                    |                         |                        |
|-------------------------------------------|--------------------|-------------------------|------------------------|
| Activity                                  | Location           | 07:00 – 23:00           | 23:00 – 07:00          |
| Resting                                   | Living Room        | 35 dB $L_{Aeq}$ 16 HOUR | --                     |
| Dining                                    | Dining Room / Area | 40 dB $L_{Aeq}$ 16 HOUR | --                     |
| Sleeping (daytime resting)                | Bedroom            | 35 dB $L_{Aeq}$ 16 HOUR | 30 dB $L_{Aeq}$ 8 HOUR |

#### 8.3.2 Calculated Internal Noise level

Calculations and assessments are therefore to be carried out in order to satisfy the above requirements of BS8233: 2014.

| External Residential Assessment | Calculated Façade Level |
|---------------------------------|-------------------------|
| Calculated Level at Residential | 38 dB(A)                |
| Partially Open Window           | 15 dB(A)                |
| Calculated internal Noise level | 23 dB(A)                |
| BS8233: 2014 criteria           | $L_{Aeq}$ 30 dB         |
| Excess Over Criteria            | -7 dB                   |
| Assessment Conclusion           | Low Impact              |

## 9 NPPF & NPSE

The National Planning Policy Framework 2024 (NPPF) and assessments to the Noise Policy Statement for England 2010 (NPSE) should be made in conjunction with each other. Paragraphs 198 - 201 of the National Planning Policy Framework 2024 (NPPF) states the following:

**Paragraph 198** Planning policies and decisions should also ensure that new development is appropriate for its location taking into account the likely effects (including cumulative effects) of pollution on health, living conditions and the natural environment, as well as the potential sensitivity of the site or the wider area to impacts that could arise from the development. In doing so they should:

- a) Mitigate and reduce to a minimum potential adverse impacts resulting from noise from new development – and avoid noise giving rise to significant adverse impacts on health and the quality of life.
- b) Identify and protect tranquil areas which have remained relatively undisturbed by noise and are prized for their recreational and amenity value for this reason.

**Paragraph 199** Planning policies and decisions should sustain and contribute towards compliance with relevant limit values or national objectives for pollutants. So far as possible these opportunities should be considered at the plan-making stage, to ensure a strategic approach and limit the need for issues to be reconsidered when determining individual applications.

**Paragraph 200** Planning policies and decisions should ensure that new development can be integrated effectively with existing businesses and community facilities (such as places of worship, pubs, music venues and sports clubs). Existing businesses and facilities should not have unreasonable restrictions placed on them as a result of development permitted after they were established. Where the operation of an existing business or community facility could have a significant adverse effect on new development (including changes of use) in its vicinity, the applicant (or 'agent of change') should be required to provide suitable mitigation before the development has been completed.

**Paragraph 201** The focus of planning policies and decisions should be on whether proposed development is an acceptable use of land, rather than the control of processes or emissions (where these are subject to separate pollution control regimes). Planning decisions should assume that these regimes will operate effectively. Equally, where a planning decision has been made on a particular development, the planning issues should not be revisited through the permitting regimes operated by pollution control authorities

**The Noise Policy Statement for England** gives various levels of effect as detailed within this report.

With the proposed development installed as detailed within this report, the development can be implemented within the guidelines of the aforementioned documents and ensure a conclusion of **NOEL – No Observed Effect Level**. This is the level below which no effect can be detected. In simple terms, below this level, there is no detectable effect on health and quality of life due to the noise.

## 10 CONCLUSIONS

### 10.1 Summary of Report

Continuous background noise measurements were undertaken between 23rd July and 28th July 2025 at the Assessment Position 1. The daytime background noise levels between 07:00 and 23:00 was found to be  $L_{A90,1 \text{ hour}}$  40 dB. This background noise level has been assessed during daytime periods only as the proposed Swimming Lessons will not operate during night time hours.

|                                                                           |                                          |                    |
|---------------------------------------------------------------------------|------------------------------------------|--------------------|
| Daytime Equivalent Existing<br>Background Noise Levels<br>(07:00 – 23:00) | Measured Levels at Assessment Position 1 | $L_{A90, t}$ 40 dB |
|---------------------------------------------------------------------------|------------------------------------------|--------------------|

### 10.2 Noise Sources

#### 10.2.1 Existing Plant Noise

The following Tables demonstrate the operational noise levels of the Swimming Lessons and the external Condenser unit. These values have been replicated onto the CADNA A 3D model to give a working scenario where suitable mitigation measures can be designed if required.

| Noise Source       | Sound Power Level |
|--------------------|-------------------|
|                    | dB(A)             |
| Fujitsu AOYG14LMCA | 67.0              |

#### 10.2.2 Internal Poolside Recorded Noise Levels

The following noise levels have been recorded within the swimming pool during a typical 3 hour lesson period from 15:00 – 18:00 hrs.

| Noise Source  | Sound Pressure Level | Octave Band Centre Frequency (Hz) |      |      |      |      |      |      |
|---------------|----------------------|-----------------------------------|------|------|------|------|------|------|
|               | dB(A)                | 63                                | 125  | 250  | 500  | 1k   | 2k   | 4k   |
| 15:00 – 16:00 | 66.5                 | 59.7                              | 54.1 | 60.4 | 63.8 | 62.1 | 59.5 | 54.1 |
| 16:00 – 17:00 | 70.7                 | 54.1                              | 52.7 | 60.9 | 66.6 | 67.2 | 64.1 | 57.4 |
| 17:00 – 18:00 | 73.2                 | 52.2                              | 49.9 | 60.1 | 68.7 | 69.7 | 66.9 | 60.6 |

**Note:** The highest of the three  $L_{Aeq, 1 \text{ HOUR}}$  dB noise levels have been used as a worst case scenario.

### 10.3 BS 4142: 2014 +A1:2019 Assessments

| Rating Level                                    | 24 dB                                | 9.1 | The acoustic feature correction is added to the specific noise level |
|-------------------------------------------------|--------------------------------------|-----|----------------------------------------------------------------------|
| Background Noise Level $L_{A90,1 \text{ hour}}$ | 40 dB                                | 8.1 | Modal Background Noise Level                                         |
| Assessment Level                                | -16 dB                               | 11  | The background level is subtracted from the rating level.            |
| Assessment                                      | -16 dB                               |     |                                                                      |
| Conclusion                                      | The assessment level is 'Low Impact' |     |                                                                      |

## 10.4 Swimming Lessons

### 10.4.1 Assessments

| External Residential Assessment | Calculated Façade Level |
|---------------------------------|-------------------------|
| Calculated Level at Residential | 38 dB                   |
| Background Noise Level          | 40 dB                   |
| Excess Over Background          | -2 dB                   |
| Assessment Conclusion           | Low Impact              |

It can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'.

### 10.4.2 Partial Open Windows

A partially open window offers a total façade attenuation of approximately 15 dB(A).

| External Residential Assessment | Calculated Façade Level        |
|---------------------------------|--------------------------------|
| Calculated Level at Residential | 38 dB(A)                       |
| Partially Open Window           | 15 dB(A)                       |
| Calculated internal Noise level | 23 dB(A)                       |
| BS8233: 2014 criteria           | L <sub>Aeq</sub> 16 Hour 30 dB |
| Excess Over Criteria            | -7 dB                          |
| Assessment Conclusion           | Low Impact                     |

## 10.5 Conclusions

The operation of the condenser and the break out noise have been assessed to establish if the swimming lessons will have a demonstrable adverse effect in terms of noise that outweigh the benefits of the development. Measurements have been undertaken in accordance with British Standard 4142: 2014 +A1:2019 and ISO 1996 – Part 2: 2017. This report has established the existing background noise levels at the closest noise sensitive façade to the site and the assessment of the impact of the site operation on nearby residential properties to the east of the site. The resulting emissions from the site running on a worst case scenario show no conflict with 'Low Impact' criteria and give a strong indication that complaint and impact on the local amenity is unlikely.

## 10.6 Conclusion Summary

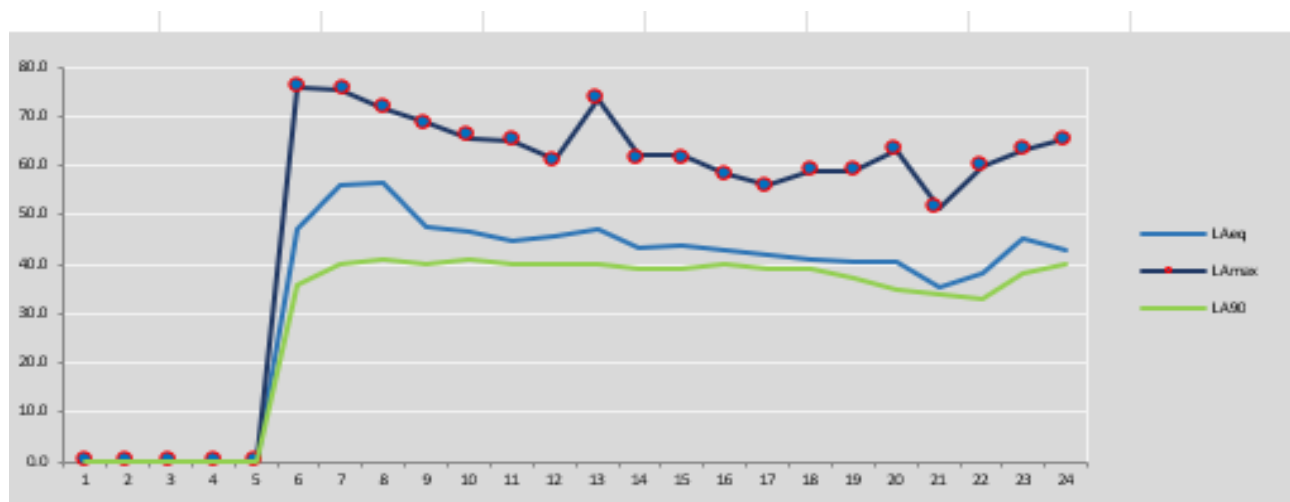
National Planning Policy Framework 2024 suggests that planning permission should be granted unless any adverse impacts of doing so would significantly and demonstrably outweigh the benefits, when assessed against the policies in the framework taken as a whole, or specific policies in the framework indicate the application should be restricted.

Based on the calculations and assessments made within this report it is the professional opinion of Impact Acoustics Ltd that the proposed development can demonstrate compliance with the National Planning Policy Framework 2024, NPPF & NPSE and that, with regards to sound, planning permission can be granted.

## 11 APPENDIX A – RESULTS

### 11.1 23<sup>rd</sup> July – 24<sup>th</sup> July 2025

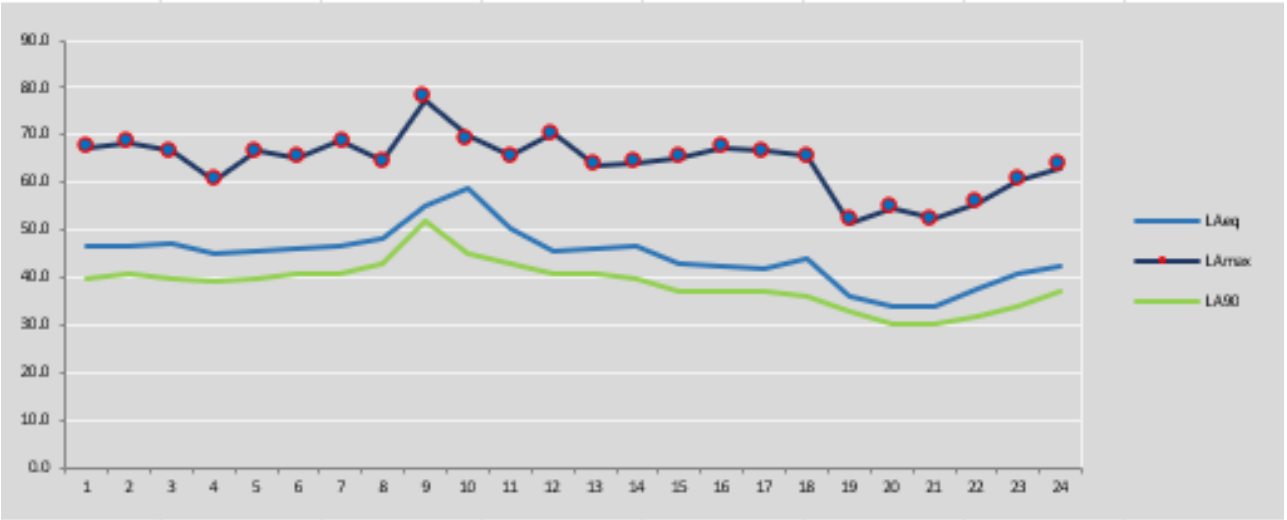
| Time                                                                                | L <sub>Max</sub> | Minimum 15 minute       |                          | 1 Hour                   |                          | Day and Night       |                           |
|-------------------------------------------------------------------------------------|------------------|-------------------------|--------------------------|--------------------------|--------------------------|---------------------|---------------------------|
| Period                                                                              |                  | L <sub>Aeq,15 min</sub> | L <sub>A90, 15 min</sub> | L <sub>Aeq, 1 hour</sub> | L <sub>A90, 1 hour</sub> | L <sub>Aeq, t</sub> | L <sub>A90, 16hours</sub> |
| 07:00 – 08:00                                                                       | 0.0              | 0.0                     | 0.0                      | 0.0                      | 0.0                      | 48.6                | 38.0                      |
| 08:00 – 09:00                                                                       | 0.0              | 0.0                     | 0.0                      | 0.0                      | 0.0                      |                     |                           |
| 09:00 – 10:00                                                                       | 0.0              | 0.0                     | 0.0                      | 0.0                      | 0.0                      |                     |                           |
| 10:00 – 11:00                                                                       | 0.0              | 0.0                     | 0.0                      | 0.0                      | 0.0                      |                     |                           |
| 11:00 – 12:00                                                                       | 0.0              | 0.0                     | 0.0                      | 0.0                      | 0.0                      |                     |                           |
| 12:00 – 13:00                                                                       | 75.9             | 47.1                    | 0.0                      | 46.9                     | 36.0                     |                     |                           |
| 13:00 – 14:00                                                                       | 75.4             | 45.3                    | 38.0                     | 56.1                     | 40.0                     |                     |                           |
| 14:00 – 15:00                                                                       | 71.5             | 49.6                    | 39.0                     | 56.3                     | 41.0                     |                     |                           |
| 15:00 – 16:00                                                                       | 68.7             | 46.4                    | 40.0                     | 47.5                     | 40.0                     |                     |                           |
| 16:00 – 17:00                                                                       | 65.5             | 45.8                    | 40.0                     | 46.4                     | 41.0                     |                     |                           |
| 17:00 – 18:00                                                                       | 64.8             | 43.6                    | 38.0                     | 44.9                     | 40.0                     |                     |                           |
| 18:00 – 19:00                                                                       | 61.0             | 44.5                    | 39.0                     | 45.6                     | 40.0                     |                     |                           |
| 19:00 – 20:00                                                                       | 73.5             | 43.8                    | 39.0                     | 47.3                     | 40.0                     |                     |                           |
| 20:00 – 21:00                                                                       | 62.0             | 41.0                    | 38.0                     | 43.1                     | 39.0                     |                     |                           |
| 21:00 – 22:00                                                                       | 62.0             | 43.1                    | 38.0                     | 43.9                     | 39.0                     |                     |                           |
| 22:00 – 23:00                                                                       | 58.5             | 41.9                    | 39.0                     | 42.9                     | 40.0                     | 42.7                | 37.0                      |
| 23:00 – 00:00                                                                       | 55.9             | 40.4                    | 38.0                     | 41.8                     | 39.0                     |                     |                           |
| 00:00 – 01:00                                                                       | 58.7             | 40.5                    | 38.0                     | 41.2                     | 39.0                     |                     |                           |
| 01:00 – 02:00                                                                       | 58.8             | 37.8                    | 36.0                     | 40.7                     | 37.0                     |                     |                           |
| 02:00 – 03:00                                                                       | 63.1             | 35.8                    | 34.0                     | 40.4                     | 35.0                     |                     |                           |
| 03:00 – 04:00                                                                       | 51.2             | 35.3                    | 33.0                     | 35.5                     | 34.0                     |                     |                           |
| 04:00 – 05:00                                                                       | 60.0             | 34.2                    | 33.0                     | 38.2                     | 33.0                     |                     |                           |
| 05:00 – 06:00                                                                       | 63.3             | 40.1                    | 36.0                     | 45.1                     | 38.0                     |                     |                           |
| 06:00 – 07:00                                                                       | 65.4             | 43.1                    | 38.0                     | 42.9                     | 40.0                     |                     |                           |
|                                                                                     |                  |                         |                          |                          |                          |                     |                           |
| Sound Pressure Levels, dB. Reference level of 2 x 10 <sup>-5</sup> Nm <sup>-2</sup> |                  |                         |                          |                          |                          |                     |                           |





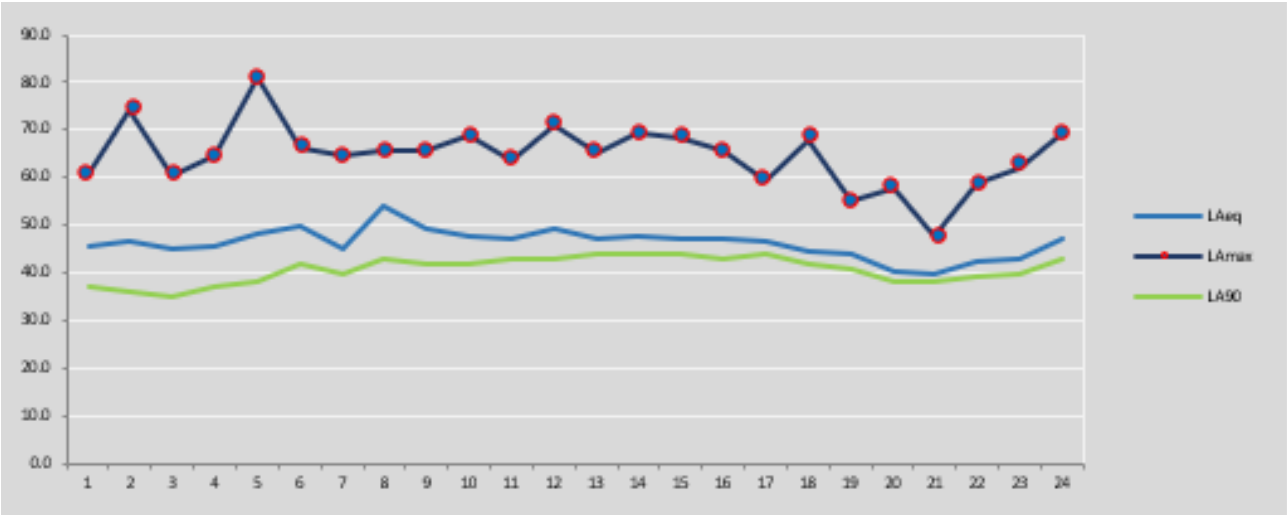
11.2 24<sup>th</sup> July – 25<sup>th</sup> July 2025

| Time                                                                                | L <sub>Max</sub> | Minimum 15 minute       |                          | 1 Hour                   |                          | Day and Night       |                           |
|-------------------------------------------------------------------------------------|------------------|-------------------------|--------------------------|--------------------------|--------------------------|---------------------|---------------------------|
| Period                                                                              |                  | L <sub>Aeq,15 min</sub> | L <sub>A90, 15 min</sub> | L <sub>Aeq, 1 hour</sub> | L <sub>A90, 1 hour</sub> | L <sub>Aeq, t</sub> | L <sub>A90, 16hours</sub> |
| 07:00 – 08:00                                                                       | 67.3             | 44.5                    | 40.0                     | 46.7                     | 40.0                     | 50.2                | 43.0                      |
| 08:00 – 09:00                                                                       | 68.6             | 45.5                    | 40.0                     | 46.5                     | 41.0                     |                     |                           |
| 09:00 – 10:00                                                                       | 66.7             | 43.8                    | 39.0                     | 47.3                     | 40.0                     |                     |                           |
| 10:00 – 11:00                                                                       | 60.5             | 43.7                    | 39.0                     | 45.1                     | 39.0                     |                     |                           |
| 11:00 – 12:00                                                                       | 66.6             | 43.4                    | 40.0                     | 45.4                     | 40.0                     |                     |                           |
| 12:00 – 13:00                                                                       | 65.2             | 44.4                    | 41.0                     | 46.2                     | 41.0                     |                     |                           |
| 13:00 – 14:00                                                                       | 68.8             | 45.7                    | 41.0                     | 46.7                     | 41.0                     |                     |                           |
| 14:00 – 15:00                                                                       | 64.4             | 47.0                    | 42.0                     | 48.2                     | 43.0                     |                     |                           |
| 15:00 – 16:00                                                                       | 77.5             | 50.1                    | 46.0                     | 55.0                     | 52.0                     |                     |                           |
| 16:00 – 17:00                                                                       | 69.7             | 50.0                    | 44.0                     | 58.6                     | 45.0                     |                     |                           |
| 17:00 – 18:00                                                                       | 65.5             | 49.5                    | 43.0                     | 50.5                     | 43.0                     |                     |                           |
| 18:00 – 19:00                                                                       | 70.5             | 44.2                    | 41.0                     | 45.3                     | 41.0                     |                     |                           |
| 19:00 – 20:00                                                                       | 63.6             | 43.6                    | 40.0                     | 46.3                     | 41.0                     |                     |                           |
| 20:00 – 21:00                                                                       | 64.1             | 42.7                    | 38.0                     | 46.5                     | 40.0                     |                     |                           |
| 21:00 – 22:00                                                                       | 65.0             | 38.7                    | 37.0                     | 42.7                     | 37.0                     |                     |                           |
| 22:00 – 23:00                                                                       | 67.3             | 38.5                    | 37.0                     | 42.2                     | 37.0                     |                     |                           |
| 23:00 – 00:00                                                                       | 66.5             | 38.7                    | 36.0                     | 41.9                     | 37.0                     |                     |                           |
| 00:00 – 01:00                                                                       | 65.9             | 38.1                    | 35.0                     | 44.0                     | 36.0                     | 42.0                | 35.0                      |
| 01:00 – 02:00                                                                       | 51.6             | 33.8                    | 33.0                     | 36.0                     | 33.0                     |                     |                           |
| 02:00 – 03:00                                                                       | 54.6             | 31.0                    | 30.0                     | 34.0                     | 30.0                     |                     |                           |
| 03:00 – 04:00                                                                       | 52.4             | 32.1                    | 29.0                     | 33.9                     | 30.0                     |                     |                           |
| 04:00 – 05:00                                                                       | 55.5             | 32.0                    | 31.0                     | 37.5                     | 32.0                     |                     |                           |
| 05:00 – 06:00                                                                       | 60.2             | 35.0                    | 32.0                     | 41.0                     | 34.0                     |                     |                           |
| 06:00 – 07:00                                                                       | 63.1             | 43.9                    | 36.0                     | 42.2                     | 37.0                     |                     |                           |
|                                                                                     |                  |                         |                          |                          |                          |                     |                           |
| Sound Pressure Levels, dB. Reference level of 2 x 10 <sup>-5</sup> Nm <sup>-2</sup> |                  |                         |                          |                          |                          |                     |                           |



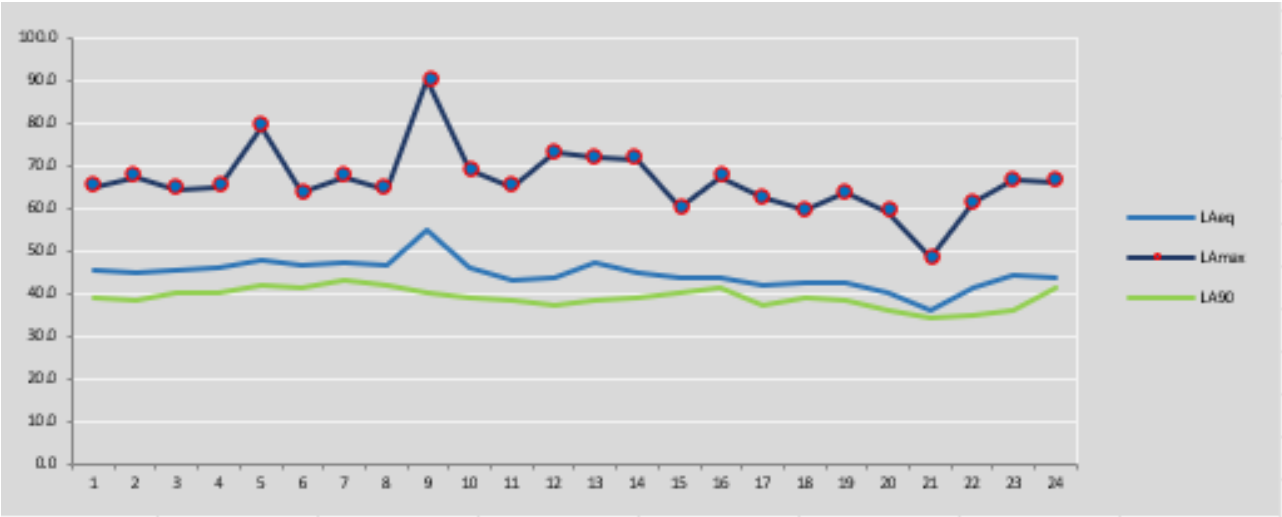
11.3 25<sup>th</sup> July – 26<sup>th</sup> July 2025

| Time                                                                                | L <sub>Max</sub> | Minimum 15 minute       |                          | 1 Hour                   |                          | Day and Night       |                           |
|-------------------------------------------------------------------------------------|------------------|-------------------------|--------------------------|--------------------------|--------------------------|---------------------|---------------------------|
| Period                                                                              |                  | L <sub>Aeq,15 min</sub> | L <sub>A90, 15 min</sub> | L <sub>Aeq, 1 hour</sub> | L <sub>A90, 1 hour</sub> | L <sub>Aeq, t</sub> | L <sub>A90, 16hours</sub> |
| 07:00 – 08:00                                                                       | 60.9             | 43.7                    | 36.0                     | 45.8                     | 37.0                     | 48.4                | 42.0                      |
| 08:00 – 09:00                                                                       | 74.0             | 43.6                    | 35.0                     | 46.8                     | 36.0                     |                     |                           |
| 09:00 – 10:00                                                                       | 60.6             | 42.2                    | 35.0                     | 44.9                     | 35.0                     |                     |                           |
| 10:00 – 11:00                                                                       | 64.8             | 44.3                    | 37.0                     | 45.8                     | 37.0                     |                     |                           |
| 11:00 – 12:00                                                                       | 81.1             | 46.5                    | 37.0                     | 48.1                     | 38.0                     |                     |                           |
| 12:00 – 13:00                                                                       | 66.3             | 43.0                    | 38.0                     | 49.6                     | 42.0                     |                     |                           |
| 13:00 – 14:00                                                                       | 64.8             | 43.0                    | 40.0                     | 44.8                     | 40.0                     |                     |                           |
| 14:00 – 15:00                                                                       | 65.5             | 48.2                    | 41.0                     | 54.1                     | 43.0                     |                     |                           |
| 15:00 – 16:00                                                                       | 65.8             | 44.4                    | 41.0                     | 49.3                     | 42.0                     |                     |                           |
| 16:00 – 17:00                                                                       | 68.7             | 45.8                    | 41.0                     | 47.9                     | 42.0                     |                     |                           |
| 17:00 – 18:00                                                                       | 63.7             | 46.3                    | 42.0                     | 47.1                     | 43.0                     |                     |                           |
| 18:00 – 19:00                                                                       | 71.2             | 47.5                    | 43.0                     | 49.4                     | 43.0                     |                     |                           |
| 19:00 – 20:00                                                                       | 65.0             | 46.6                    | 44.0                     | 47.2                     | 44.0                     |                     |                           |
| 20:00 – 21:00                                                                       | 69.5             | 46.8                    | 43.0                     | 47.4                     | 44.0                     |                     |                           |
| 21:00 – 22:00                                                                       | 68.4             | 46.4                    | 44.0                     | 47.4                     | 44.0                     |                     |                           |
| 22:00 – 23:00                                                                       | 65.7             | 45.6                    | 43.0                     | 47.0                     | 43.0                     |                     |                           |
| 23:00 – 00:00                                                                       | 59.5             | 45.3                    | 43.0                     | 46.4                     | 44.0                     | 43.9                | 41.0                      |
| 00:00 – 01:00                                                                       | 67.9             | 43.9                    | 41.0                     | 44.6                     | 42.0                     |                     |                           |
| 01:00 – 02:00                                                                       | 54.9             | 42.7                    | 40.0                     | 43.8                     | 41.0                     |                     |                           |
| 02:00 – 03:00                                                                       | 57.8             | 38.8                    | 37.0                     | 40.4                     | 38.0                     |                     |                           |
| 03:00 – 04:00                                                                       | 47.6             | 39.5                    | 38.0                     | 39.8                     | 38.0                     |                     |                           |
| 04:00 – 05:00                                                                       | 59.0             | 40.3                    | 38.0                     | 42.2                     | 39.0                     |                     |                           |
| 05:00 – 06:00                                                                       | 62.2             | 42.2                    | 39.0                     | 43.0                     | 40.0                     |                     |                           |
| 06:00 – 07:00                                                                       | 69.5             | 46.0                    | 40.0                     | 47.0                     | 43.0                     |                     |                           |
|                                                                                     |                  |                         |                          |                          |                          |                     |                           |
| Sound Pressure Levels, dB. Reference level of 2 x 10 <sup>-5</sup> Nm <sup>-2</sup> |                  |                         |                          |                          |                          |                     |                           |



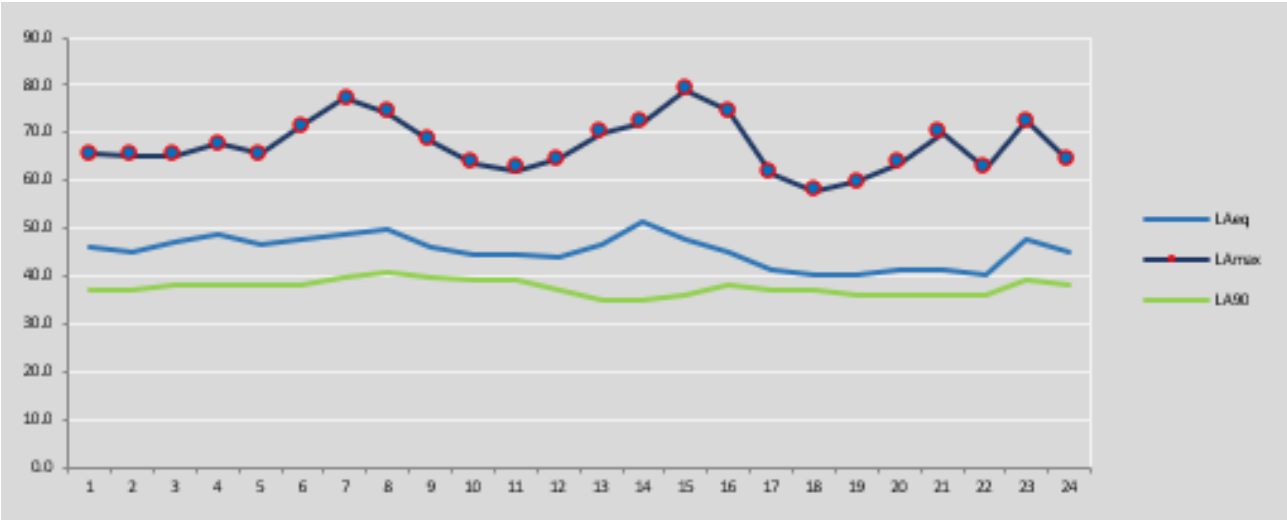
11.4 26<sup>th</sup> July – 27<sup>th</sup> July 2025

| Time                                                                                | L <sub>Max</sub> | Minimum 15 minute       |                          | 1 Hour                   |                          | Day and Night       |                           |
|-------------------------------------------------------------------------------------|------------------|-------------------------|--------------------------|--------------------------|--------------------------|---------------------|---------------------------|
| Period                                                                              |                  | L <sub>Aeq,15 min</sub> | L <sub>A90, 15 min</sub> | L <sub>Aeq, 1 hour</sub> | L <sub>A90, 1 hour</sub> | L <sub>Aeq, t</sub> | L <sub>A90, 16hours</sub> |
| 07:00 – 08:00                                                                       | 65.0             | 43.7                    | 38.0                     | 45.1                     | 39.0                     | 47.1                | 40.0                      |
| 08:00 – 09:00                                                                       | 67.2             | 42.3                    | 38.0                     | 44.6                     | 38.0                     |                     |                           |
| 09:00 – 10:00                                                                       | 64.2             | 43.2                    | 39.0                     | 45.1                     | 40.0                     |                     |                           |
| 10:00 – 11:00                                                                       | 64.9             | 43.6                    | 39.0                     | 45.7                     | 40.0                     |                     |                           |
| 11:00 – 12:00                                                                       | 79.1             | 44.8                    | 42.0                     | 47.7                     | 42.0                     |                     |                           |
| 12:00 – 13:00                                                                       | 63.4             | 44.5                    | 40.0                     | 46.2                     | 41.0                     |                     |                           |
| 13:00 – 14:00                                                                       | 67.0             | 45.8                    | 42.0                     | 47.3                     | 43.0                     |                     |                           |
| 14:00 – 15:00                                                                       | 64.2             | 45.5                    | 41.0                     | 46.8                     | 42.0                     |                     |                           |
| 15:00 – 16:00                                                                       | 90.2             | 47.1                    | 39.0                     | 54.7                     | 40.0                     |                     |                           |
| 16:00 – 17:00                                                                       | 68.7             | 41.7                    | 39.0                     | 45.9                     | 39.0                     |                     |                           |
| 17:00 – 18:00                                                                       | 64.8             | 40.6                    | 37.0                     | 43.1                     | 38.0                     |                     |                           |
| 18:00 – 19:00                                                                       | 73.1             | 40.0                    | 37.0                     | 43.4                     | 37.0                     |                     |                           |
| 19:00 – 20:00                                                                       | 72.0             | 42.4                    | 37.0                     | 46.9                     | 38.0                     |                     |                           |
| 20:00 – 21:00                                                                       | 71.5             | 42.1                    | 38.0                     | 44.5                     | 39.0                     |                     |                           |
| 21:00 – 22:00                                                                       | 60.3             | 41.6                    | 39.0                     | 43.4                     | 40.0                     |                     |                           |
| 22:00 – 23:00                                                                       | 67.4             | 42.3                    | 40.0                     | 43.4                     | 41.0                     |                     |                           |
| 23:00 – 00:00                                                                       | 62.3             | 39.4                    | 37.0                     | 42.0                     | 37.0                     | 42.4                | 37.0                      |
| 00:00 – 01:00                                                                       | 59.3             | 41.5                    | 38.0                     | 42.3                     | 39.0                     |                     |                           |
| 01:00 – 02:00                                                                       | 63.6             | 39.8                    | 37.0                     | 42.5                     | 38.0                     |                     |                           |
| 02:00 – 03:00                                                                       | 58.6             | 36.3                    | 35.0                     | 39.8                     | 36.0                     |                     |                           |
| 03:00 – 04:00                                                                       | 48.4             | 35.2                    | 34.0                     | 36.0                     | 34.0                     |                     |                           |
| 04:00 – 05:00                                                                       | 61.2             | 36.7                    | 35.0                     | 40.9                     | 35.0                     |                     |                           |
| 05:00 – 06:00                                                                       | 66.3             | 37.2                    | 35.0                     | 44.0                     | 36.0                     |                     |                           |
| 06:00 – 07:00                                                                       | 65.9             | 44.8                    | 36.0                     | 43.4                     | 41.0                     |                     |                           |
|                                                                                     |                  |                         |                          |                          |                          |                     |                           |
| Sound Pressure Levels, dB. Reference level of 2 x 10 <sup>-5</sup> Nm <sup>-2</sup> |                  |                         |                          |                          |                          |                     |                           |



11.5 27<sup>th</sup> July – 28<sup>th</sup> July 2025

| Time                                                                                | L <sub>Max</sub> | Minimum 15 minute       |                          | 1 Hour                   |                          | Day and Night       |                           |
|-------------------------------------------------------------------------------------|------------------|-------------------------|--------------------------|--------------------------|--------------------------|---------------------|---------------------------|
| Period                                                                              |                  | L <sub>Aeq,15 min</sub> | L <sub>A90, 15 min</sub> | L <sub>Aeq, 1 hour</sub> | L <sub>A90, 1 hour</sub> | L <sub>Aeq, t</sub> | L <sub>A90, 16hours</sub> |
| 07:00 – 08:00                                                                       | 65.7             | 43.1                    | 36.0                     | 45.8                     | 37.0                     | 47.4                | 38.0                      |
| 08:00 – 09:00                                                                       | 65.2             | 40.6                    | 35.0                     | 45.0                     | 37.0                     |                     |                           |
| 09:00 – 10:00                                                                       | 65.0             | 46.5                    | 37.0                     | 47.2                     | 38.0                     |                     |                           |
| 10:00 – 11:00                                                                       | 67.7             | 47.3                    | 37.0                     | 48.8                     | 38.0                     |                     |                           |
| 11:00 – 12:00                                                                       | 65.8             | 45.3                    | 37.0                     | 46.8                     | 38.0                     |                     |                           |
| 12:00 – 13:00                                                                       | 71.6             | 44.7                    | 38.0                     | 47.5                     | 38.0                     |                     |                           |
| 13:00 – 14:00                                                                       | 77.3             | 46.6                    | 39.0                     | 49.0                     | 40.0                     |                     |                           |
| 14:00 – 15:00                                                                       | 74.3             | 48.0                    | 40.0                     | 49.7                     | 41.0                     |                     |                           |
| 15:00 – 16:00                                                                       | 68.3             | 44.3                    | 39.0                     | 46.0                     | 40.0                     |                     |                           |
| 16:00 – 17:00                                                                       | 63.8             | 44.1                    | 39.0                     | 44.6                     | 39.0                     |                     |                           |
| 17:00 – 18:00                                                                       | 62.2             | 43.6                    | 38.0                     | 44.4                     | 39.0                     |                     |                           |
| 18:00 – 19:00                                                                       | 64.5             | 41.3                    | 36.0                     | 44.2                     | 37.0                     |                     |                           |
| 19:00 – 20:00                                                                       | 69.9             | 39.7                    | 34.0                     | 46.4                     | 35.0                     |                     |                           |
| 20:00 – 21:00                                                                       | 72.0             | 47.3                    | 35.0                     | 51.5                     | 35.0                     |                     |                           |
| 21:00 – 22:00                                                                       | 79.2             | 40.6                    | 35.0                     | 47.7                     | 36.0                     |                     |                           |
| 22:00 – 23:00                                                                       | 74.6             | 39.6                    | 36.0                     | 45.0                     | 38.0                     |                     |                           |
| 23:00 – 00:00                                                                       | 61.7             | 40.0                    | 37.0                     | 41.5                     | 37.0                     | 43.5                | 38.0                      |
| 00:00 – 01:00                                                                       | 57.5             | 39.5                    | 37.0                     | 40.1                     | 37.0                     |                     |                           |
| 01:00 – 02:00                                                                       | 59.8             | 37.9                    | 34.0                     | 40.2                     | 36.0                     |                     |                           |
| 02:00 – 03:00                                                                       | 63.4             | 38.3                    | 36.0                     | 41.2                     | 36.0                     |                     |                           |
| 03:00 – 04:00                                                                       | 70.1             | 38.6                    | 36.0                     | 41.4                     | 36.0                     |                     |                           |
| 04:00 – 05:00                                                                       | 62.6             | 37.2                    | 35.0                     | 40.1                     | 36.0                     |                     |                           |
| 05:00 – 06:00                                                                       | 72.4             | 41.5                    | 38.0                     | 47.9                     | 39.0                     |                     |                           |
| 06:00 – 07:00                                                                       | 64.2             | 46.1                    | 42.0                     | 45.0                     | 38.0                     |                     |                           |
|                                                                                     |                  |                         |                          |                          |                          |                     |                           |
| Sound Pressure Levels, dB. Reference level of 2 x 10 <sup>-5</sup> Nm <sup>-2</sup> |                  |                         |                          |                          |                          |                     |                           |



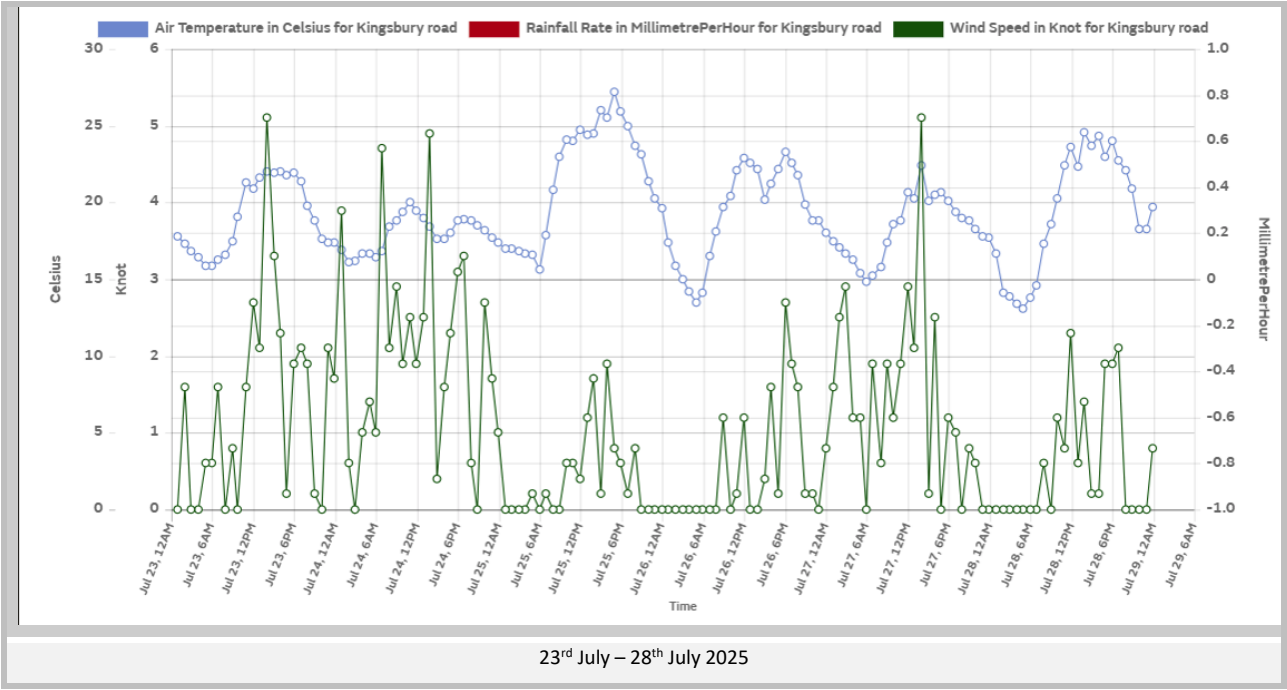
## 12 APPENDIX B - UNCERTAINTY

In line with the requirements of section 10 of British Standard 4142: 2014 it is expected that the reported expanded uncertainty with a confidence limit of 95% and assuming a convergence of  $k=2$  is likely to be  $\pm 2.1$ dB. Uncertainty, in this instance has been minimised by undertaking longer background noise measurements over a 120-hour period.

### 12.1 Uncertainty Budget

| Sources of Uncertainty                                                                                                                                                                                                                                                                                                         | Uncertainty Notes               | Commentary                    | Value (half Width) | Convert to Same Units (dB) | Distribution Divisor |                    |               | Standard Uncertainty (u) dB |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|--------------------|----------------------------|----------------------|--------------------|---------------|-----------------------------|
|                                                                                                                                                                                                                                                                                                                                |                                 |                               |                    |                            | Normal               | Rectangular        | Other         |                             |
| Measurement Position                                                                                                                                                                                                                                                                                                           | Choice of position, ,           | 1m in 20m                     |                    | 0.2                        |                      | rect( $\sqrt{3}$ ) |               | 0.13                        |
|                                                                                                                                                                                                                                                                                                                                | microphone orientation          | Type 1 0 - 30deg              | 0.4                |                            | Normal               |                    |               | 0.50                        |
| Instrumentation                                                                                                                                                                                                                                                                                                                | Calibration                     | Calibration Drift             | 0.1                |                            | Normal               |                    |               | 0.10                        |
|                                                                                                                                                                                                                                                                                                                                | Accuracy and precision (type 1) | Type 1 practical              | 1.9                |                            |                      | rect( $\sqrt{3}$ ) |               | 0.50                        |
| Background Noise Level                                                                                                                                                                                                                                                                                                         | Timing of Measurement           |                               |                    |                            |                      |                    |               |                             |
| Background Noise Level                                                                                                                                                                                                                                                                                                         | Modal Analysis Day              | Calculated Standard Deviation | 3.7                |                            |                      |                    | s/ $\sqrt{n}$ | 0.39                        |
| Background Noise Level                                                                                                                                                                                                                                                                                                         | Modal Analysis Night            | Calculated Standard Deviation | 2.2                |                            |                      |                    | s/ $\sqrt{n}$ | 0.16                        |
| Combined Uncertainty (root sum of squares)                                                                                                                                                                                                                                                                                     |                                 |                               |                    |                            |                      |                    |               | 1.05                        |
| Expanded Uncertainty $U = Kuc$ (95% Confidence $K=2$ )                                                                                                                                                                                                                                                                         |                                 |                               |                    |                            |                      |                    |               | 2.1                         |
| Final Answer Expressed as Value $\pm U$ dB with a confidence Level of 95%                                                                                                                                                                                                                                                      |                                 |                               |                    |                            |                      |                    |               |                             |
| REFERENCE: Uncertainty Budget Calculated in line with M3003: The Expression of Uncertainty and Confidence in Measurement Edition 3, November 2012 and A Good Practice Guide on the Sources and Magnitude of Uncertainty Arising in the Practical Measurement of Environmental Noise N J Craven, G Kerry Edition 1a – May 2007. |                                 |                               |                    |                            |                      |                    |               |                             |

13 APPENDIX C – WEATHER CONDITIONS



## 14 APPENDIX D – BS 4142:2019+A1:2019 INFORMATION

In accordance with BS 4142:2014:A1:2019, section 12, the following information is provided.

### 14.1 Measurement Information – Location 1

| Details & Justification            | Measurement position 1 was located at the rear of the existing premises, representative of the Nearest Noise Sensitive Premises |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Height from Ground                 | 1.5m                                                                                                                            |
| Reflective Surfaces                | None                                                                                                                            |
| Distance to NSP                    | 15m                                                                                                                             |
| Distance to Source                 | 10m                                                                                                                             |
| Topographical Change (m)           | None                                                                                                                            |
| Ground between Source and Receiver | 100% Soft                                                                                                                       |
| Windshield Information             | Norsonic Outdoor Microphone Protection Kit Model No: 1212                                                                       |
| Distance from Source to Receiver   | 15m                                                                                                                             |
| NSP* - Noise Sensitive Premises    |                                                                                                                                 |

# END OF REPORT #